

MM

Mapalo Musonda

FULL STACK DEVELOPER

CONTACT

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Lusaka, Zambia

LINKS

github.com/Penza2024

LinkedIn

Portfolio

CORE SKILLS

TypeScript / JavaScript

React / React Native

Node.js / Express

Supabase / PostgreSQL

Docker / CI-CD

Python / C++ / Java

TECHNOLOGIES

React

TypeScript

React Native

Expo

Tailwind CSS

Node.js

Vite

Docker

Kubernetes

GitHub Actions

Supabase

PostgreSQL

Firebase

ESP32

LoRaWAN

MQTT

Three.js

PHP

MySQL

Git

AI TOOLS

Claude

ChatGPT

Cursor

CERTIFICATES

MetaTrader 5 Algorithmic Trading

Coursera · Jan 2024

Huawei AI and Applications

Huawei Talent Online · 2025

Kaggle: Intro to Deep Learning

Kaggle · 2024

Intro to IoT & Digital Transformation

Cisco Networking Academy · May 2026

PROFILE

Full-stack developer graduating BSc Computer Science in August 2026, with hands-on experience across mobile, web, IoT, and DevOps. I build production systems end-to-end — from React Native apps with live payment integrations to ESP32 sensor networks over LoRaWAN — and have delivered professional client work alongside open source contributions. Strong in TypeScript, React, Docker, and Supabase, with AI tools as a core part of my workflow.

EXPERIENCE & PROJECTS

Smart Grid Monitoring System

2025 – 2026

Final Year Project · ZESCO, Zambia's national electricity utility

End-to-end energy monitoring platform: C++ RTOS firmware on Heltec ESP32 nodes (LoRa SX1262, multi-constellation GPS, PZEM-004T meter, OLED, OTA), cloud sync to Supabase, and dual React dashboards for grid operators and consumers.

React

TypeScript

Supabase

C++ / RTOS

ESP32

LoRaWAN

GPS

Reservit — Event Management Platform

2024 – 2025

Mobile application · React Native / Expo

Event ticketing app with venue dashboards, real-time booking, QR ticket generation, and Airtel Money payment processing secured via Supabase Edge Functions.

React Native

TypeScript

Supabase

Airtel Money

BusTickTeT — Bus Ticketing App

2024

Mobile application · React Native / Expo

Multi-company bus management with seat booking, trip scheduling, and an admin compliance dashboard with audit logging.

React Native

TypeScript

Supabase

Company Document Management System

2024 – 2025

Full-stack web application · CodeIgniter 4 / PHP

Secure document platform with role-based access control, version control, department hierarchies, and a calendar system. Containerised across development, staging, and production environments with PHPUnit and property-based testing.

CodeIgniter 4

PHP

MySQL

Redis

Docker

PHPUnit

Portfolio Website — mapalo.dev

2024 – Present

Personal project · Live on Vercel

Responsive portfolio with interactive 3D WebGL rendering (Three.js), particle animation, dark/light theme, and a full blog and publications section. Built with HTML5, Tailwind CSS, and vanilla JavaScript.

HTML5

Tailwind CSS

JavaScript

Three.js

Vercel

Client Work (NDA)

2024 – 2025

Professional engagements · Zambian software firms

Full-stack enterprise platform and a freight logistics web app with containerised deployment and CI/CD pipeline for two separate Zambian clients.

Docker

CI/CD

Full Stack

OPEN SOURCE & RESEARCH

RuView — contributor

ESP32 / C++ / Python · WiFi CSI sensing

Firmware and DSP pipeline contributions to a WiFi presence-detection platform using Channel State Information.

ZICTA Research — 2025

Accepted publication

"AI-Powered Language Translation to Boost Tourism in Zambia" — accepted by ZICTA 2025.

EDUCATION

BSc. Computer Science

